

Where Should We Begin? A Low-Level Exploration of Weight Initialization Impact on Quantized Behaviour of Deep Neural Networks

Stone Yun
Alexander Wong
Email: {s22yun, a28wong}@uwaterloo.ca

University of Waterloo
University of Waterloo

Abstract

With the proliferation of deep convolutional neural network (CNN) algorithms for mobile processing, limited precision quantization has become an essential tool for CNN efficiency. Consequently, various works have sought to design fixed precision quantization algorithms and quantization-focused optimization techniques that minimize quantization induced performance degradation. However, there is little concrete understanding of how various CNN design decisions/best practices affect quantized inference behaviour. Weight initialization strategies are often associated with solving issues such as vanishing/exploding gradients but an often-overlooked aspect is their impact on the final trained distributions of each layer. We present an in-depth, fine-grained ablation study of the effect of different weights initializations on the final distributions of weights and activations of different CNN architectures. The fine-grained, layer-wise analysis enables us to gain deep insights on how initial weights distributions will affect final accuracy and quantized behaviour. To our best knowledge, we are the first to perform such a low-level, in-depth quantitative analysis of weights initialization and its effect on quantized behaviour.

Introduction

Deep Convolutional Neural Networks (CNN) have enabled dramatic advances in the field of computer vision. An explosion of recent research has demonstrated incredible results in applications such as image classification [1–3], object detection [4–7], image segmentation [8, 9] and many more. As CNNs have become an increasing part of everyday life, their usage has expanded to mobile processing. With this expansion comes issues of minimizing latency, area, and power. Fixed point quantization of CNN weights and activations has become an essential tool for running efficient CNN inference. Various works have explored different quantization algorithms [10–12] to minimize the loss of information when mapping the weights and activations of a CNN into a discretized space. With these methods, the weights and activations can be represented as an n -bit (most often 8-bit) integer rather than 32-bit floating point numbers. Consequently, simple integer multiply-accumulate (MAC) operations can be performed rather than costly floating point arithmetic. This leads to significant savings in both computation (integer arithmetic) and storage (typically 8-bits per value or less).

Mobile hardware accelerators are usually limited in the types of operations that can be massively parallelized for fast execution. Thus, more complex quantization methods are often not supported by existing hardware. As such, other works have focused on quantization-algorithm-specific optimization methods (Eg. targeting 8-bit uniform quantization). These include quantization-aware fine-tuning [10] and differential optimization of quantization parameters [13, 14], eg. finding the optimal max/min thresholds of each weight/activation tensor for minimal quantized degradation. These methods train a model that is robust to quantized perturbations by simulating the error/noise of fixed point arithmetic.

Despite significant works demonstrating ways to recover near floating-point performance using reduced precision inference, there is still little understanding of how different design decisions can affect the quantized inference behaviour of CNNs. Weight initialization strategies are often designed with the goal of solving issues such as vanishing/exploding gradient [15–17]. However, an often-overlooked aspect of weights initialization is its impact on the final trained distributions of each layer. As they determine our starting point on the loss surface, initial distributions of each weight tensor will have a profound impact on the final trained model. Gradient descent is an incremental process with many small, noisy steps. Thus, an intelligent weight initialization strategy will have significant impact on the local minima reached on our path through the loss space. With regards to quantization, this means that weight initialization choices will have significant impact on the dynamic ranges of the weights and activations in a trained CNN. Thus, affecting the noise in our system and the expected quantized inference behaviour.

We propose a framework for in-depth, fine-grained quantitative analysis of the impact of various weights initialization strategies on final accuracy and quantized behaviour. By analyzing the trained distributions of each layer's weights and activations, we can gain deep insight on how different weights initialization strategies will affect the dynamic ranges of each layer. This in turn provides insight on the quantized behaviour of a CNN. Furthermore, we analyze the effect of these different weights initializations for a small set of different CNN architectures. Thus, we are able to isolate and observe the interplay between the CNN architecture choices (the parameterization) and the weights initialization strategy (the starting point on the parameterized loss surface). To our best knowledge, we are the first to perform such a systematic, low-level, quantitative analysis of weights initialization strategies and quantized behaviour. Furthermore, our framework for fine-grained analysis is applicable to analyzing any number of CNN design choices such as layer types, batch size, learning rate schedule etc.

2 Background

In early research, neural network parameters were often randomly initialized based on sampling from a normal or uniform distribution. The respective variance and range of these distributions would be hyperparameters for the practitioner to decide. While easily taken for granted, several works such as [15–17] have provided rigorous mathematical proofs showing how intelligent weights initialization strategies can solve issues of vanishing and exploding gradients. These works define fan_in and fan_out of a fully connected layer as the input/output units respectively. For convolution, it is defined as Eq. 1 where K is the kernel width (assume square kernel). They provide mathematical proofs on their proposed fan_in/fan_out -aware initialization strategies that scale the variance of gradients at each layer. Thus, avoiding failure modes created by vanishing and exploding gradients. While the introduction of Batch Normalization (BatchNorm) [18] layers has greatly mitigated training issues involving gradient scales, the choice of "where to begin" in the parameterized loss space is still extremely relevant. An often-overlooked effect of these initialization strategies is their impact on the trained dynamic ranges of each layer. As gradient descent is a noisy, iterative process with small, incremental steps, the final dynamic ranges of each layer are profoundly impacted by their starting point.

$$fan_{in/out} = K \times K \times channel_{in/out} \quad (1)$$

3 Fine-grained Layerwise Analysis

Besides a high-level study of how different weight initializations affect 32-bit floating point (fp32) and eight-bit quantized (quint8) accuracy, we also wish to gain detailed insight on the layer-wise distributions of final trained weights and activations. This information can give us an in-depth look at how the learning dynamics of various weight initializations play out. Furthermore, the dynamic ranges of each weight/activation tensor determine the resolution of the quantized step-size and, by extension, the quantization noise in a CNN. Thus, this analysis can help explain the observed quantized inference behaviour of different trained models. We propose systematically ablating through a variety of different weight initialization strategies while tracking the dynamic ranges of each layer's weights and activations during training. In this way, we can isolate the effect of these different design choices and analyze the changing distributions at each layer. We also track the "average channel precision". Average channel precision is defined as Eq. 2. Channel precision in this context is the ratio between an individual channel's range and the range of the entire layer. [19] uses this precision quantity to algorithmically maximize the channel precisions of each layer in a network prior to quantization. It can be seen as a measure of how well the overall layer-wise quantization encodings represent the information in each channel.

$$average_precision = \frac{1}{K} \sum_{i=1}^K \frac{range_i}{range_{tensor}} \quad (2)$$